

ISE 555: Unit 2 Assignment

Fall 2026

For this assignment, please complete at least **4 out of the first 6 problems**. The writing assignment (Problem 7) is mandatory. A PDF file containing all your solutions as well as the writing assignment must be submitted on Brightspace by 4:00pm on Monday, September 28th. Make sure you have access to your submission during class on the day that the assignment is due. This will be helpful when presenting your solutions to the class in person.

- 1) Consider a feasible region S defined by a set of linear constraints as follows:

$$S = \{x: Ax \leq b\}$$

Prove that S is convex.

- 2) Express $(2,2)^T$ as a convex combination of $(0,0)^T$, $(1,4)^T$, and $(3,1)^T$

- 3) For each of the following functions, determine if it is convex, concave, both, or neither on the real line. Note that “both” can mean convex for some values of x and concave for others.

(a) $f(x) = 3x^2 + 4x - 5$

(b) $f(x) = 4 - 5x + 3x^2$

(c) $f(x) = \exp(x^2)$

(d) $f(x) = 7x - 15$

- 4) Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & f(x) = -x_1 - x_2 \\ \text{subject to} & x_1 + x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{array}$$

- (a) Determine the feasible directions at $x = (0, 0)^T$, $(0, 1)^T$, $(1, 1)^T$, and $(0, 2)^T$.
- (b) Which (if any) of the points can be local minimizers.

- 5) Find all descent directions for the linear function $f(x) = x_1 - 2x_2 + 3x_3$. Does your answer depend on the value of x ?
- 6) Consider the set defined by the constraints $x_1 + x_2 = 1$, $x_1 \geq 0$, and $x_2 \geq 0$. At each of the following point determine the set of feasible directions:
- (a) $x = (0, 1)^T$
 - (b) $x = (1, 0)^T$
 - (c) $x = (0.5, 0.5)^T$
- 7) WRITING ASSIGNMENT: Suppose you work for a manufacturing company and they are adding one more machine to a large factory of 84 machines. You had previously optimized the scheduled use of the 84 machines. You are now asked to optimize the scheduled use of the 85 machines. Explain in writing to your manager why a local optimization procedure may be better to use than a global optimization procedure. *[This only needs to be a paragraph – not a long essay]*